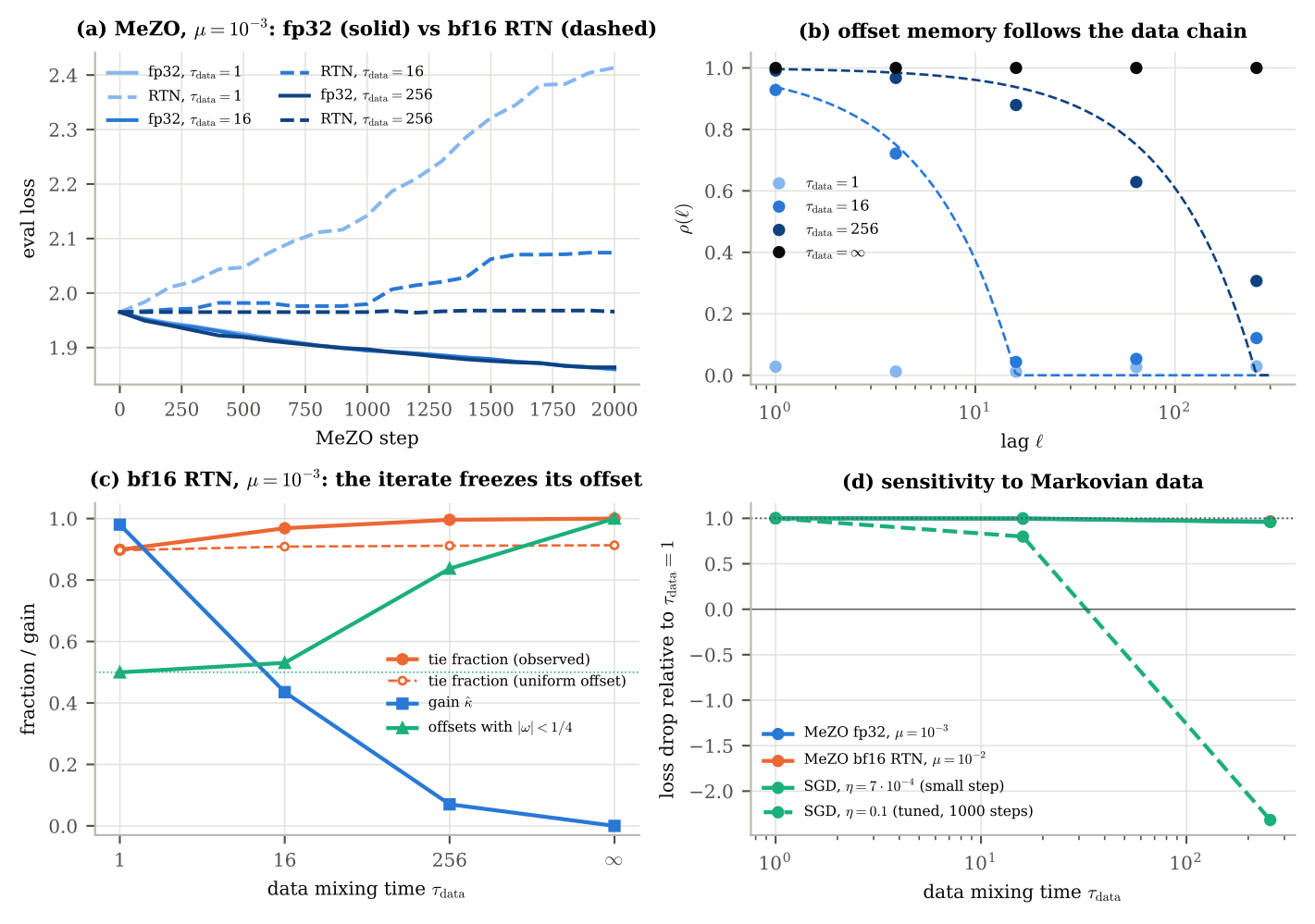

(a) MeZO, $\mu = 10^{-3}$: fp32 (solid) vs bf16 RTN (dashed)
fp32, $\tau_{\text{data}} = 1$
RTN, $\tau_{\text{data}} = 1$
fp32, $\tau_{\text{data}} = 16$
RTN, $\tau_{\text{data}} = 16$
fp32, $\tau_{\text{data}} = 256$
RTN, $\tau_{\text{data}} = 256$
eval loss
MeZO step
(b) offset memory follows the data chain
$\tau_{\text{data}} = 1$
$\tau_{\text{data}} = 16$
$\tau_{\text{data}} = 256$
$\tau_{\text{data}} = \infty$
$\rho(\ell)$
lag ℓ
(c) bf16 RTN, $\mu = 10^{-3}$: the iterate freezes its offset
tie fraction (observed)
tie fraction (uniform offset)
gain $\hat{\kappa}$
offsets with $|\omega| < 1/4$
fraction / gain
data mixing time τ_{data}
(d) sensitivity to Markovian data
MeZO fp32, $\mu = 10^{-3}$
MeZO bf16 RTN, $\mu = 10^{-2}$
SGD, $\eta = 7 \cdot 10^{-4}$ (small step)
SGD, $\eta = 0.1$ (tuned, 1000 steps)
loss drop relative to $\tau_{\text{data}} = 1$
data mixing time τ_{data}